



sTMS Interface Functional Specification

Singapore Polytechnic – iCRM System

Version 1.0

Document Version

Version	Effective Date	Summary of Changes	Created/ Updated By
0.1	26/09/2022	The initial version of SP iCRM System interface functional specification	AvePoint
0.2	05/10/2022	Revised base on SP email feedback. • 2.1 Removed MOE_Student file. • 2.2 Updated interface frequency and timing. • 3.1 Revised the length of field.	AvePoint
0.3	12/10/2022	Revised base on SP email feedback. • 2.1 Updated the diagram. • 3.2 Removed MOE_Student feed file.	AvePoint
0.4	20/10/2022	Revised data field length.	AvePoint
0.5	25/10/2022	Updated data length of company name, street name, and student Id.	AvePoint
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Document Acceptance Form

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1. Introduction

Purpose of this document

This Interface Functional Specification (IFS) is to provide the agreed baseline of communication requirements and detailed technical specification between the SaaS shared Training Management System (sTMS) and Singapore Polytechnic iCRM System. sTMS required to transfer CET current term students, MOE students, graduates, course, race information to SP iCRM System.

sTMS shall identify student, course, race information based on the UAM Student ID, enrolment status, graduation information and personal information to generate corresponding TXT files. sTMS shall generate and deposit these TXT files encrypted by PGP protocol in the sTMS SFTP server accordingly. Singapore Polytechnic can use the transferred TXT files for marketing purpose.

Intended Audience

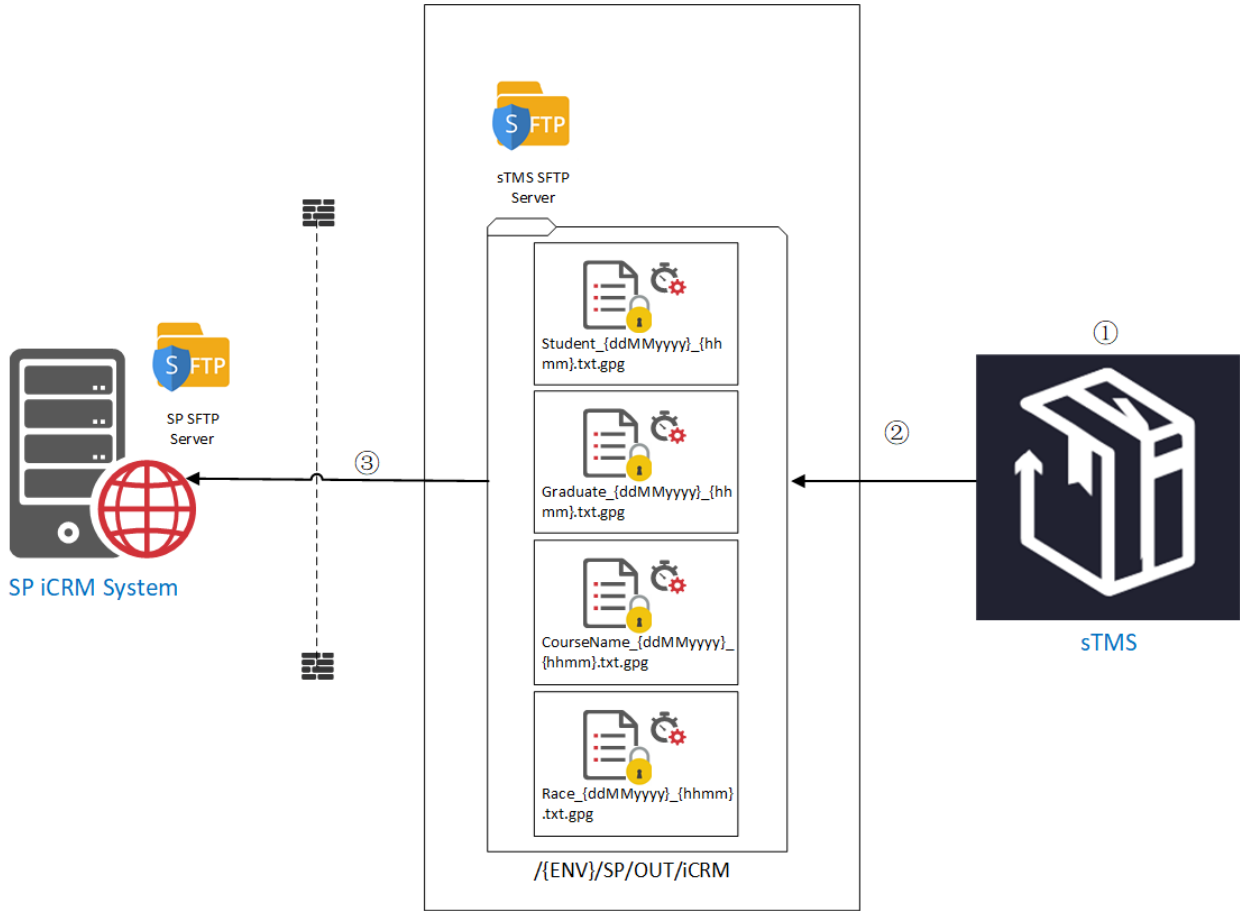
This technical document is used for system communication. Therefore, the intended audiences are the technical stakeholders of sTMS and SP iCRM System.

2. Technical Overview

System Flow

There are 3 roles involved in the interface diagram.

- **sTMS**: shared Training Management System.
- **sTMS SFTP Server**: A SFTP server hosted in Azure that stores encrypted files used for data exchange.
- **SP iCRM System**: Singapore Polytechnic iCRM system.



SN	Description
1	sTMS generates the TXT files based on student's enrollment status, current term, course category and course type on a pre-defined schedule.
2	sTMS configures timer job to push TXT files into sTMS SFTP server after encryption with PGP protocol.
3	sTMS SFTP server push TXT files into SP SFTP server.

Interface Design Summary

Output Interface file

Interface File Name	Student_{ddMMyyyy}_{hhmm}.txt.gpg Graduate_{ddMMyyyy}_{hhmm}.txt.gpg CourseName_{ddMMyyyy}_{hhmm}.txt.gpg Race_{ddMMyyyy}_{hhmm}.txt.gpg
Data Source System	shared Training Management System
Data Destination System	Singapore Polytechnic iCRM System
Method	SFTP
File Path	/ {ENV} / SP / OUT / iCRM ENV: UAT, PROD
Encryption Method	PGP
Frequency	Daily
Timing	Start dumping files to sTMS SFTP server at 6:30 PM, push files to SP SFTP server at 8:00 PM
Folder Retention Policy	Files under the / {ENV} / SP / OUT / iCRM are retained for a maximum of 30 days in the sTMS SFTP server, after this period the files will be automatically deleted.
File Transfer Policy	sTMS shall transfer empty files with header info if there are no records should be sent to SP iCRM System.
Possible Exception and Exception Handling	File loss or transfer failure due to sTMS internal exception. E.g., Timer service down, etc. Exception handling steps: • sTMS support team shall be informed by email of the transfer failure. • sTMS support team checks the relevant logs to troubleshoot the cause of the failure. • If the issue is due to the transfer file could not being found in SP SFTP server: • sTMS support team will manually rerun the timer job to trigger file generation and deposit the file in SP SFTP server.



Business Rule

The following information in the table would be synced daily:

No.	Table	Syncing Frequency	Scope
1	Student	Incremental data	All active SP student data within current term for last 7 days.
2	Graduate	Full-load data	SP MOE course graduated students' data with last 3 terms (Including current term)
3	CourseName	Full-load data	The courses information contained in the "Student" and "Graduate" table.
4	Race	Full-load data	Race information contained in the "Graduate" table.

3. Interface Fields

Interface Field

Student

iCRM Field Name	sTMS Field Name	Description	Data Type	Remarks
Application Id	Application ID	Application ID of student	String(\$nvarchar(37))	Sample: Intake ID-12345
Student ID	Student ID	UAM Common ID of student	String(\$nvarchar(8))	7 digitals+ 1 checksum, Sample: 2212345S
Salutation	-	-	-	Default to Blank
Full Name	Name	Name of student	String(\$nvarchar(66))	Sample: Bruce Li
Gender	Gender	Gender of student	String(\$nvarchar(1))	Value list: F, M, U
Residential Status	Citizenship type	Citizenship type of student	String(\$nvarchar(1))	Mapping refer to 4.1 Residential Status Sample: A, C, P
iChat Email	UAM email	UAM email address of student	String(\$nvarchar(255))	Sample: EMAIL@ABC.com
Email - Personal Email/Home	Email	Personal email of student	String(\$nvarchar(255))	Sample: bruce@gmail.com
Mobile Phone	Mobile Number	Mobile number of student	String(\$nvarchar(20))	Sample: +65 91241234 +86 13578781212
Home Phone	-	-	-	Default to Blank
NRIC	NRIC/FIN	NRIC/FIN number of student	String(\$nvarchar(9))	Sample: S1234567G
Passport No	Passport No	Passport number of student	String(\$nvarchar(9))	Sample: E12345678
Academic Status	Enrollment Status	Course intake enrollment status of student	String(\$nvarchar(25))	Mapping refer to 4.2 Academic Status Sample: Activate

School (Academic School)	School Code (Course Level)	Course level school code	String(\$nvarchar(4))	Sample: SHSS
Course Unique ID	Course Code	Course code	String(\$nvarchar(24))	Sample: SP-SE0001
Course Type	Course Category	Course type and course category	String(\$nvarchar(4))	Mapping refer to 4.3 Course Type Sample: C300
Academic Course	Course short name	Course short name	String(\$nvarchar(255))	Sample: SE

Graduate and MOE_Student:

iCRM Field Name	sTMS Field Name	Description	Data Type	Remarks
Student Id	Student ID	UAM Common ID of student	String(\$nvarchar(8))	7 digitals + Checksum, Sample: 2212345S
Salutation	-	-	-	Default to Blank
Student Class	Course intake No.	Course intake number	String(\$nvarchar(255))	Sample value: SP-CEM101-220001
Application Exercise	-	-	-	Default to CET
Last Institution (Pre-SP)	Award Institution	The institution that awards the highest educational qualifications	String(\$nvarchar(255))	Sample: ITE
Secondary School	SecondarySchoolName	The Institution name when educational qualification type is Secondary	String(\$nvarchar(255))	Sample: Queensway Secondary School
Pre-SP Qualification	-	-	-	Default to Blank
Mobile Phone	Mobile Number	Mobile number of student	String(\$nvarchar(20))	Sample: +65 91241234 +86 13578781212
Email	Email	Personal email of student	String(\$nvarchar(255))	Sample: bruce@gmail.com

Alternate Email	-	-	-	Default to Blank
Employment Status	Employment Status	Course intake enrollment status of student	String(\$nvarchar(25))	Mapping refer to 4.2 Academic Status Sample: Activate
Citizenship	Citizenship Type	Citizenship type of student	String(\$nvarchar(1))	Mapping refer to 4.1 Residential Status Sample: A, C, P
Date of Birth	Date of birth	Date of birth of student	String(\$DD/MM/YYYY)	Sample: 20/01/1995
Country of Birth	Country of birth	Country of birth of student	String(\$nvarchar(50))	Value list refer to 4.4 Country Name Sample: Australia
Marital Status	Marital status	Marital status of student	String(\$nvarchar(1))	Have value when student apply bursary Value list refer to 4.7 Marital Status
Race	Race	Race of student	String(\$nvarchar(2))	Value list refer to 4.5 Race Name Sample: SG, CN
Religion	-	-	-	Default to Blank
Name of Employer	Name of employer	Name of employer of the latest work experience of the student	String(\$nvarchar(100))	Sample: AvePoint
Floor & Unit No	Floor No. - Unit No.	#Floor No. - Unit No. of residential address	String(\$nvarchar(10))	Sample: #02-11
Block Number	Block/Building number	Block/Building number of residential address	String(\$nvarchar(10))	Sample: BLK123
Street Name	Street name	Street name of residential address	String(\$nvarchar(32))	Sample: Street A
Building Name	Building name	Building name of residential address	String(\$nvarchar(66))	Sample: Building A
Country	Country code	Country code of residential address	String(\$nvarchar(50))	Value list refer to 4.4 Country Name Sample: Singapore
Zip / Postal Code	Postal code	Postal code of residential address	String(\$nvarchar(6))	Sample: 123321



GPA	cGPA	Cumulative GPA of student	String(\$nvarchar(10))	Sample: 3.8
Graduating Year	Graduated-BOE date	Date of BOE	String(\$nvarchar(4))	2022
School (Academic School)	School Code (Course Level)	School code	String(\$nvarchar(4))	Sample value: SOBM
Course Name	Course Name	Course name	String(\$nvarchar(255))	Sample value: Big Data Analytics
Course Type	Course Category	Course type and course category	String(\$nvarchar(4))	Mapping refer to 4.3 Course Type Sample: C300
NRIC	NRIC/FIN No.	NRIC/FIN number of graduate student	String(\$nvarchar(9))	Sample value: S1234567C
Passport No	Passport No.	Passport number of graduate student	String(\$nvarchar(9))	Sample value: E12345678
Gender	Gender	Gender of graduate student	String(\$nvarchar(1))	Value list: M, F, U
Full name	Name	Full name of graduate student	String(\$nvarchar(66))	Sample value: Jonah Louis
Home Phone	-	-	-	Default to Blank
Business Phone	-	-	-	Default to Blank

CourseName

iCRM Field Name	sTMS Field Name	Description	Data Type	Remarks
Unique Name	Course Name	Course name	String(\$nvarchar(255))	Sample value: Big Data Analytics
Unique Code	Course Code	Course code	String(\$nvarchar(24))	Sample value: SP-123456
School	School Code	School code of the course	String(\$nvarchar(4))	Sample value: LSC



Qualification Type	Course Category	Course type and course category	String(\$nvarchar(4))	Mapping refer to 4.6 Qualification Type Sample: Part Time
Course Type	Course Category	Course type and course category	String(\$nvarchar(4))	Mapping refer to 4.3 Course Type Sample: C300
Status	-	-	-	Default to Active

Race

iCRM Field Name	sTMS Field Name	Description	Data Type	Remarks
Unique Name	Race	Race name	String(\$nvarchar(20))	Sample value: CHINESE
Unique Code	Race code	Race code	String(\$nvarchar(2))	Sample value: CN
Status	-	-	-	Default to Active

Sample Files


Student_16122021_0105.txt


Graduate_16122021_0105.txt


CourseName_16122021_0105.txt


Race_16122021_0105.txt

4. Appendix

For Residential status, sTMS will follow SG-DRM and send status code to SP in below table SP value list.

4.1 Residential Status

sTMS Value List	SP Value List – Status code
Foreigner	A
Singapore citizen	C
Singapore PR	P

4.2 Academic Status

sTMS Value List	SP Value List
Enrolled	Activate
Completed	Completion of Program
Deferred	Defer Enrollment
Suspended	Suspension
Transferred	Transfer to Other Career
Withdrawn	Administrative Withdrawal

4.3 Course Type

MOE Indicator	sTMS Course Category	sTMS Course Type	SP Course Type
MOE	Part-time Diploma (PTD)	Academic Full Qualification (FQ)	C300
		Academic Certificate (MC)	C300
		Academic Modular (SBMC)	C300
	Specialist Diploma (SD)	Academic Full Qualification (FQ)	C302
		Academic Certificate (PDC)	C302
		Academic Modular (SBMC)	C302
	Advanced Diploma (AD)	Academic Full Qualification (FQ)	C301
		Academic Certificate (PDC)	C301
		Academic Modular (SBMC)	C301
	Diploma Conversion	Academic Full Qualification (FQ)	C303
		Academic Certificate (PDC)	C303
		Academic Modular (SBMC)	C303
	MOE Academic Courses	Part-Time Nitec	C300
		Part-Time Higher Nitec	C300

		ITE Skills Certificate (ISC)	C300
	Certifiable Courses	SkillsFuture Series	C200
		SkillsFuture Digital Confidence	C200
		SkillsFuture Digital Workplace	C200
		SkillsFuture Career Transition Programme (SCTP)	C200
		SGUnited Skills (SGUS) Programme	C200
		SGUnited Mid-Career Pathways (SGUP) Programme	C200
		Career Conversion Programme	C200
		Work-Learn Bootcamp	C200
		Certificate of Competency (COC) - SkillsFuture Series	C200
		General Education (GE)	C200
		Joint Industry Certificate (JIND)	C200
		Train The Trainer (TTT)	C200
		Workforce Skills Qualification (WSQ) Full Qualification	C200
		Workforce Skills Qualification (WSQ) Modular Course	C200
		Certifiable Short Courses	C100
	Non-Certifiable Courses	SGUnited Skills (SGUS) Programme	C200
		Work-Learn Bootcamp	C200
		Non-Certifiable Short Courses	C100
		SkillsFuture Digital Workplace	C200
		Seminars and Webinars	C200
	Customised Programme	Certifiable customised Courses	C200
		Non-Certifiable customised Courses	C200
MOE	Work-Study Programmes	Work-Study Certificates Leading To Part-Time Diploma	C300
		Work-Study Post-Diploma Specialist Diploma	C302
		Work-Study Post-Diploma Leading To Specialist Diploma	C302
		Work-Study Post-Diploma Advanced Diploma	C301
		Workforce Skills Qualification (WSQ)	C200
MOE		Work-Study Post-Diploma Diploma Conversion	C303
		Work Study Certificate (WSC)	C200

4.4 Country Name

For country name, sTMS follow MyInfo country list and will send the sTMS value to SP with capitalize the first letter of each word and leave the other letters lowercase.



Country Name.xlsx

4.5 Race Name

For race name and code, sTMS will follow MyInfo race list and send sTMS value and code to SP, SP can store the race information in iCRM accordingly.



Race Name.xlsx

4.6 Qualification Type

MOE Indicator	sTMS Course Category	sTMS Course Type	SP Qualification Type
MOE	Part-time Diploma (PTD)	Academic Full Qualification (FQ)	Part Time
		Academic Certificate (MC)	Part Time
		Academic Modular (SBMC)	Part Time
	Specialist Diploma (SD)	Academic Full Qualification (FQ)	Specialist Diploma
		Academic Certificate (PDC)	Specialist Diploma
		Academic Modular (SBMC)	Specialist Diploma
	Advanced Diploma (AD)	Academic Full Qualification (FQ)	Advanced Diploma
		Academic Certificate (PDC)	Advanced Diploma
		Academic Modular (SBMC)	Advanced Diploma
	Diploma Conversion	Academic Full Qualification (FQ)	Diploma (Conversion)
		Academic Certificate (PDC)	Diploma (Conversion)
		Academic Modular (SBMC)	Diploma (Conversion)
	MOE Academic Courses	Part-Time Nitec	Regular Diploma
		Part-Time Higher Nitec	Regular Diploma
		ITE Skills Certificate (ISC)	Regular Diploma
	Certifiable Courses	SkillsFuture Series	Certification/Qualification
		SkillsFuture Digital Confidence	Certification/Qualification
		SkillsFuture Digital Workplace	Certification/Qualification
		SkillsFuture Career Transition Programme (SCTP)	Certification/Qualification
		SGUnited Skills (SGUS) Programme	SGUS Programme
		SGUnited Mid-Career Pathways (SGUP) Programme	Certification/Qualification

		Career Conversion Programme	Certification/ Qualification
		Work-Learn Bootcamp	Certification/ Qualification
		Certificate of Competency (COC) - SkillsFuture Series	Certification/ Qualification
		General Education (GE)	Certification/ Qualification
		Joint Industry Certificate (JIND)	Certification/ Qualification
		Train The Trainer (TTT)	Certification/ Qualification
		Workforce Skills Qualification (WSQ) Full Qualification	WSQ Course
		Workforce Skills Qualification (WSQ) Modular Course	WSQ Course
		Certifiable Short Courses	Short Course
	Non-Certifiable Courses	SGUnited Skills (SGUS) Programme	SGUS Programme
		Work-Learn Bootcamp	Certification/ Qualification
		Non-Certifiable Short Courses	Short Course
		SkillsFuture Digital Workplace	Certification/ Qualification
		Seminars and Webinars	Certification/ Qualification
	Customised Programme	Certifiable customised Courses	Certification/ Qualification
		Non-Certifiable customised Courses	Certification/ Qualification
MOE	Work-Study Programmes	Work-Study Certificates Leading To Part-Time Diploma	Part Time
		Work-Study Post-Diploma Specialist Diploma	Post Diploma
		Work-Study Post-Diploma Leading To Specialist Diploma	Post Diploma
		Work-Study Post-Diploma Advanced Diploma	Post Diploma
MOE		Workforce Skills Qualification (WSQ)	WSQ Course
		Work-Study Post-Diploma Diploma Conversion	Post Diploma
		Work Study Certificate (WSC)	Certification/ Qualification



4.7 Marital Status

For Marital status, sTMS will follow SG-DRM and send status code to SP in below table SP value list.

sTMS Value List	SP Value List – Marital status code
Single	1
Married	2
Widowed	3
Divorced	5